

NOTEBOOK

OVERVIEW

ZJU-CHINA2026





Notebook — GAG Retention Module

CPC / Link-N protein expression, purification and GAG-interaction assays

Timeline of the Work

Date (2026)	Work performed	Record
22–23 July	Plasmid reconstitution, transformation into BL21(DE3), recovery of transformants and preparation of DH5α glycerol stocks	1
24 July–12 August	Initial CPC expression screening, repeated SDS-PAGE / Western blot checks, and unsuccessful small- and large-scale purification attempts	2
13–27 August	Review of the initial construct design; CPC coding-sequence redesign and codon optimisation; arrival of the revised plasmids and transformation	3
28 August–1 September	Small-scale expression screening of CPC, Link-N and CPC+Link-N at 16 °C / 30 °C and 9 h / 16 h	4
2–6 September	Scale-up expression, His-tag affinity purification, SDS-PAGE analysis and BCA concentration measurement	5
4–12 September	Establishment and optimisation of the CS-A agarose electrophoretic mobility assay; repeat tests with SUMO-containing and cleaved protein samples	6
7–15 September	ULP1-mediated SUMO removal, reverse affinity purification, ultrafiltration and protein concentration checks	7
11 September	DogTag-mediated ligation check using the GFP-SnoopTag reporter and SnoopLigase2	8
12 and 25 September	AviTag biotinylation trials, streptavidin-binding validation and preparation of protein-coated magnetic beads	9
27–28 September	DMMB measurement of residual CS-A in bead-assay supernatants and preliminary statistical analysis	10

1 Plasmid reconstitution, transformation and starter cultures

Date: 22–23 July 2026

Procedures

1. The plasmid DNA supplied in dried form was dissolved in sterile water to a recorded working concentration of 100 ng/µL, aliquoted and stored at −20 °C.
2. The plasmids were introduced into chemically competent E. coli BL21(DE3) for protein expression.
3. Single colonies from the DH5α and BL21 transformation plates were picked separately and inoculated into LB broth containing kanamycin. Cultures were grown at 37 °C and 200 rpm overnight to provide starter cultures.
4. DH5α cultures were preserved as glycerol stocks by mixing culture with 50% glycerol at a 1:1 ratio, dispensing 1 mL aliquots and storing them at −80 °C.

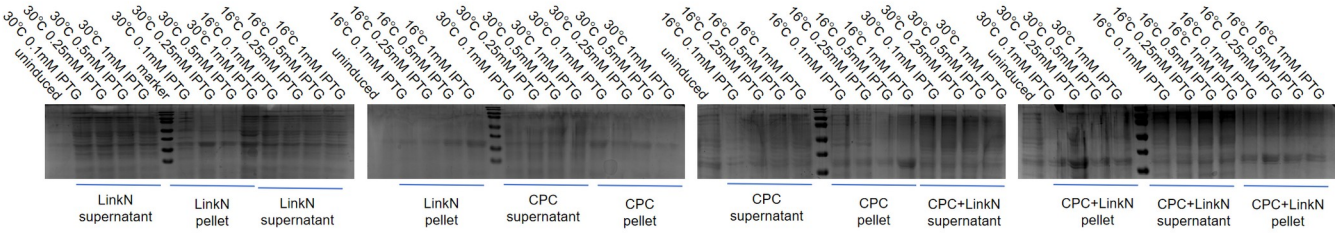
Note. Colonies were available for subsequent induction trials. The working kanamycin concentration recorded for LB preparation was 50 µg/mL.

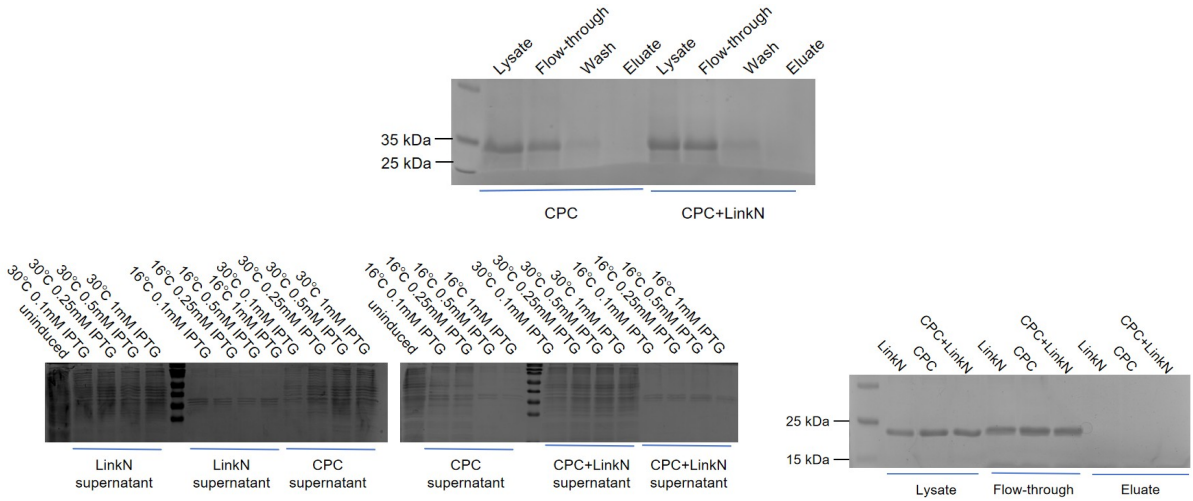
2 Initial CPC expression screening and troubleshooting

Date: 24 July–12 August 2026

Procedures

1. Starter cultures were diluted 1:50 into fresh medium and expanded until the culture reached the recorded induction point. A 1 mL uninduced sample was retained for comparison.
2. The first induction screen tested IPTG concentrations of 0.1, 0.25, 0.5 and 1.0 mM under two conditions: 16 °C overnight and 30 °C for 4 h, giving eight screening conditions.
3. Additional induction trials were subsequently run at 37 °C for 4 h and at 30 °C for 9 h or 16 h. These extensions were intended to determine whether the weak target signal could be improved by changing temperature or duration.
4. For the 25 July harvest, samples were centrifuged at 4 °C and 4,000 rpm for 10 min. Pellets were resuspended in PBS, recentrifuged at 4 °C and 12,000 rpm for 2 min, aliquoted and stored at −20 °C. For soluble/insoluble analysis, cells were sonicated using a recorded 5 s on / 5 s off cycle for 6 min, then centrifuged at 4 °C and 12,000 rpm for 20 min to separate supernatant and pellet fractions.
5. SDS-PAGE was repeated after the first gels were poorly resolved. Small-scale His-tag affinity purification was attempted from 26–30 July, with lysate, flow-through and eluate samples examined by SDS-PAGE. Large-scale induction and purification attempts continued into early August; Western blotting was added after bands were observed in nominally uninduced samples.





Note. The early SDS-PAGE images were difficult to interpret because of weak or diffuse bands and an over-destained gel. In subsequent purification attempts, the lysate and flow-through contained protein, but the eluate was nearly blank or showed no clear enrichment of the target. A band was also observed in an uninduced sample during the Western blot checks, raising concern about basal expression or a background signal. The combined evidence did not establish a reliable expression-and-recovery condition for the initial repetitive CPC design. The records support a troubleshooting decision, but not a definitive assignment of the uninduced band to the target protein.

3 CPC coding-sequence redesign and preparation of revised constructs

Date: 13–27 August 2026

Procedures

1. After the initial expression attempts, the CPC design was reconsidered. A Gibson-based reconstruction was discussed and primers were drafted, but the plan was revisited after the team rechecked the vector backbone and received additional results.
2. The CPC coding sequence was then redesigned using the literature-supported CPC+14A sequence RRRR(AARRR)$_3$R. Codon optimisation for E. coli was repeated while reducing DNA-level repetitiveness in the coding sequence.
3. The final construct set comprised pET-28a(+)-CPC (5,827 bp), pET-28a(+)-LinkN (5,805 bp) and pET-28a(+)-CPC+LinkN (5,887 bp). The constructs shared the same expression-vector framework and retained AviTag for biotinylation and DogTag for ligation assays. CPC+LinkN contained both peptide segments separated by a linker.
4. LB medium was prepared for the renewed transformation and induction work. The recorded liquid-medium formulation per litre was 10 g tryptone, 5 g yeast extract and 10 g NaCl; agar was added for solid medium. Kanamycin was used at 50 µg/mL.
5. The final plasmid in the set arrived on 27 August. The plasmids were transformed, plated on selective medium and taken forward for expression screening. A DogTag-containing ligation-system protein was also recovered for later reporter-assay controls.

Note. The revised sequence design produced detectable CPC-containing protein in the subsequent expression screen.

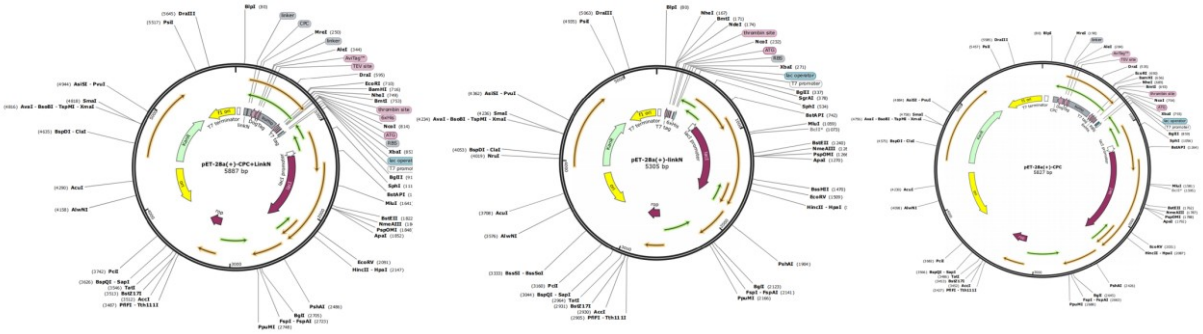
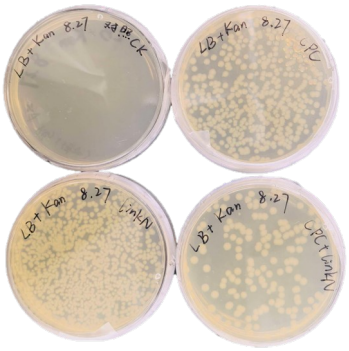


Figure 1. Plasmid maps of pET-28a(+)-CPC, pET-28a(+)-LinkN and pET-28a(+)-CPC+LinkN. The maps and lengths are reproduced from the GAG Retention Module Results document.



4 Small-scale expression screening of the revised CPC / Link-N constructs

Date: 28 August–1 September 2026

Procedures

1. Transformants were picked and grown overnight at 37 °C and 220 rpm. For small-scale expression, CPC, Link-N and CPC+Link-N cultures were inoculated at approximately 1:100 into 10 mL medium.
2. Each construct was tested at 0.5 mM IPTG under four induction conditions: 30 °C for 9 h, 30 °C for 16 h, 16 °C for 9 h and 16 °C for 16 h. The three constructs and four conditions yielded 12 groups.
3. Expression samples were analysed by SDS-PAGE. Western blotting was used to help distinguish the target proteins when the Coomassie-stained bands were weak or difficult to assign.

Note. After the coding-sequence redesign, target bands became detectable and Western blotting supported the presence of Link-N, CPC and CPC+Link-N proteins. Basal signal was still present in at least one nominally uninduced sample, so the expression system was not completely free of leakage. The working condition carried forward was 30 °C with 0.5 mM IPTG for 16 h.

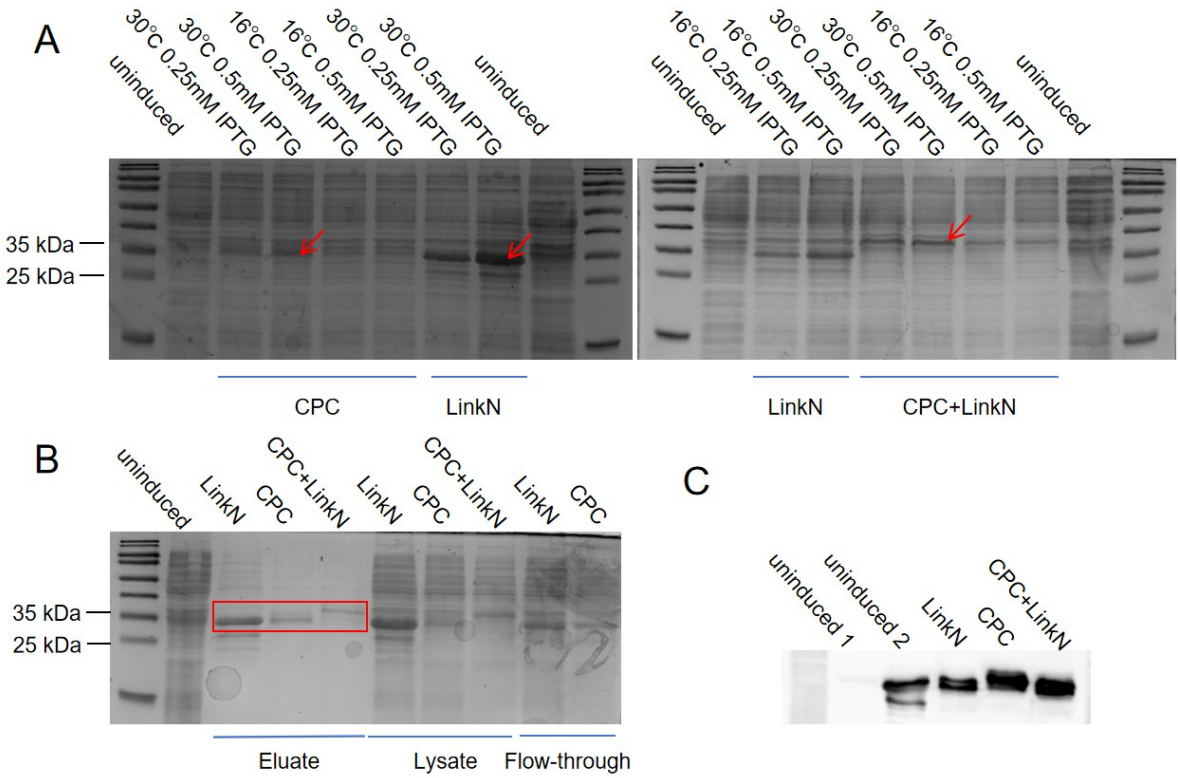


Figure 2. SDS-PAGE and Western blot analysis of the revised CPC, Link-N and CPC+Link-N expression system. The red arrows mark the target-band positions used in the expression assessment; an uninduced signal was also noted in the Western blot panel.

5 Scale-up expression, His-tag affinity purification and BCA quantification

Date: 2–6 September 2026

Procedures

1. The selected expression condition was used for larger-scale production of CPC, Link-N and CPC+Link-N. Cells were harvested and lysed; clarified lysates were applied to His-tag affinity purification.
2. Samples from the purification workflow, including lysate, flow-through, wash and eluate fractions, were retained and compared by SDS-PAGE. A later gel record specifically labels the uninduced sample, lysate, flow-through, wash and eluate for the three constructs.
3. Eluate fractions were concentrated and buffer-exchanged using 0.5 mL centrifugal ultrafiltration devices with a 5 kDa molecular-weight cut-off. Protein concentration was measured by BCA after 100-fold sample dilution, as recorded in the laboratory notes.

Note. the following concentrations for the SUMO-fusion preparations before the later cleavage workflow: SUMO-CPC, 0.43 mg/mL; SUMO-CPC-Link-N, 0.26 mg/mL; and SUMO-Link-N, 17.51 mg/mL.

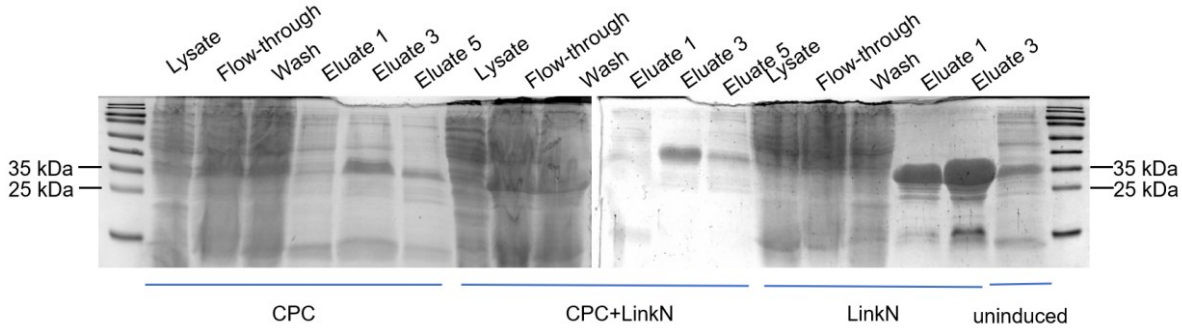


Figure 3. SDS-PAGE analysis of fractions collected during scale-up His-tag affinity purification of CPC, CPC+Link-N and Link-N constructs.

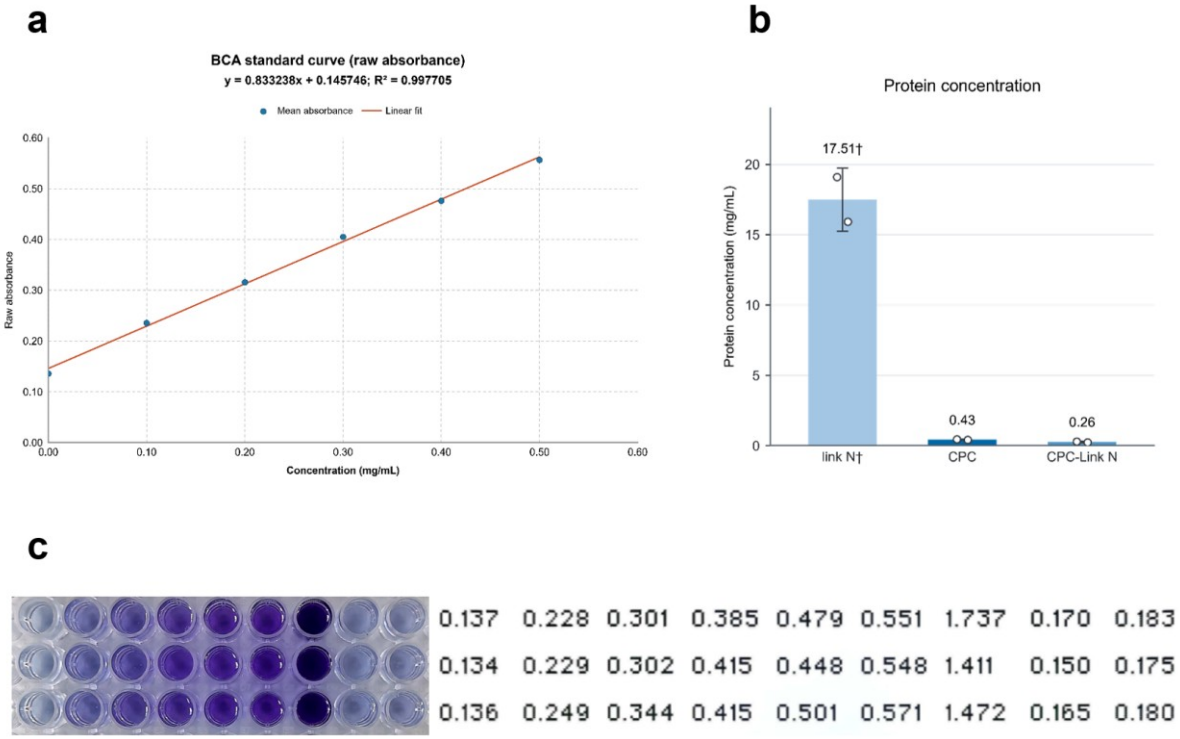


Figure 4. BCA quantification of the SUMO-fusion proteins before cleavage. The plot reports 0.43 mg/mL for SUMO-CPC, 0.26 mg/mL for SUMO-CPC-Link-N and 17.51 mg/mL for SUMO-Link-N; the standard-curve fit is shown in panel A.

6 Establishment and optimisation of the CS-A agarose mobility assay

Date: 4–12 September 2026

Procedures

1. The first assay compared CS-A alone with CS-A mixed with SUMO-CPC, SUMO-CPC-Link-N or SUMO-Link-N. The protein concentration series was recorded in the Results summary as 0.5, 1, 2.5, 5 and 10 µM, giving a 16-lane assay including the CS-A-only lane.
2. The initial electrophoresis run lasted approximately 50 min, with bromophenol blue used as the tracking dye. CS-A migrated faster than the dye front and travelled too far for reliable comparison at the end of the run.
3. CS-A-only samples were then tested at different input amounts. Clear CS-A bands were detectable, with the band intensity changing with the amount loaded. Based on this check, the electrophoresis duration was shortened to approximately 30–40 min.

4. The mixed-protein samples were rerun after the timing change. A further attempt was made using protein samples after SUMO cleavage, but no interpretable bands were obtained in that run. Additional gels were attempted on 10–12 September.

Note. The assay was able to detect CS-A when it was loaded alone, but neither the original fusion proteins nor the subsequently processed samples gave a reproducible, concentration-dependent mobility shift. The result is therefore a negative or inconclusive readout from this assay, not proof that CPC lacks GAG-binding activity.

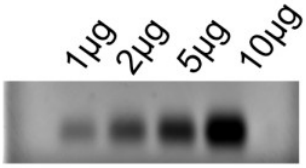
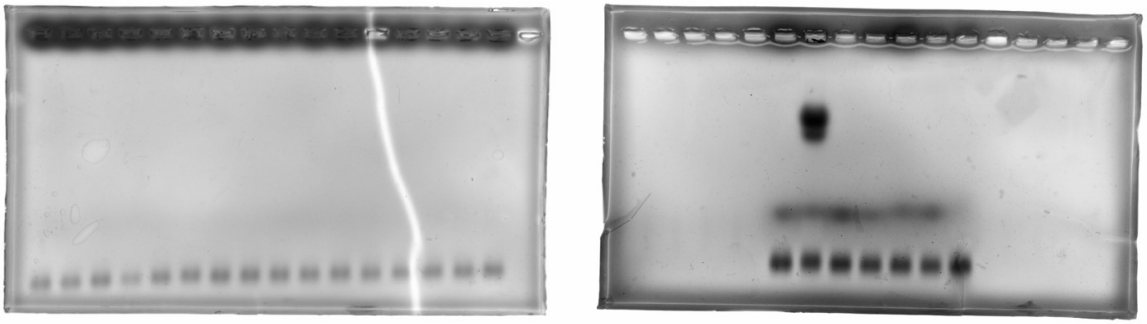


Figure 5. CS-A-only control in the optimised agarose electrophoresis assay. CS-A was detectable at input amounts of 1, 2, 5 and 10 µg, supporting the suitability of the staining workflow for visualising the GAG substrate.



7 ULP1-mediated SUMO removal, reverse affinity purification and protein recovery

Date: 7–15 September 2026

Procedures

1. SUMO removal was first tested at small scale and then repeated with larger protein preparations. Samples taken before and after ULP1 treatment were analysed by SDS-PAGE to assess the appearance of cleavage products.
2. Following digestion, reverse His-tag affinity purification was used to remove ULP1 and other His-tagged components. The recovered fractions were analysed by SDS-PAGE, followed by further ultrafiltration and concentration measurement by BCA.
3. A cleavage product was visible for the processed proteins, but uncleaved material remained in the samples, indicating that digestion was incomplete. The small-scale cleaved material used for a subsequent CS-A gel attempt did not produce a usable electrophoretic readout.

Note. The Results summary contains concentration values from more than one point in the workflow. Values reported immediately after ULP1 cleavage / reverse affinity purification were Link-N 0.30 mg/mL, CPC 0.21 mg/mL and CPC-Link-N 0.19 mg/mL. A later post-ultrafiltration summary lists Link-N 1.42 mg/mL, CPC 0.14 mg/mL and CPC-Link-N 0.69 mg/mL. The dated 15 September notebook entry records LinkN 1.41 mg/mL, CPC 0.70 mg/mL and CPC-LinkN 0.15 mg/mL.

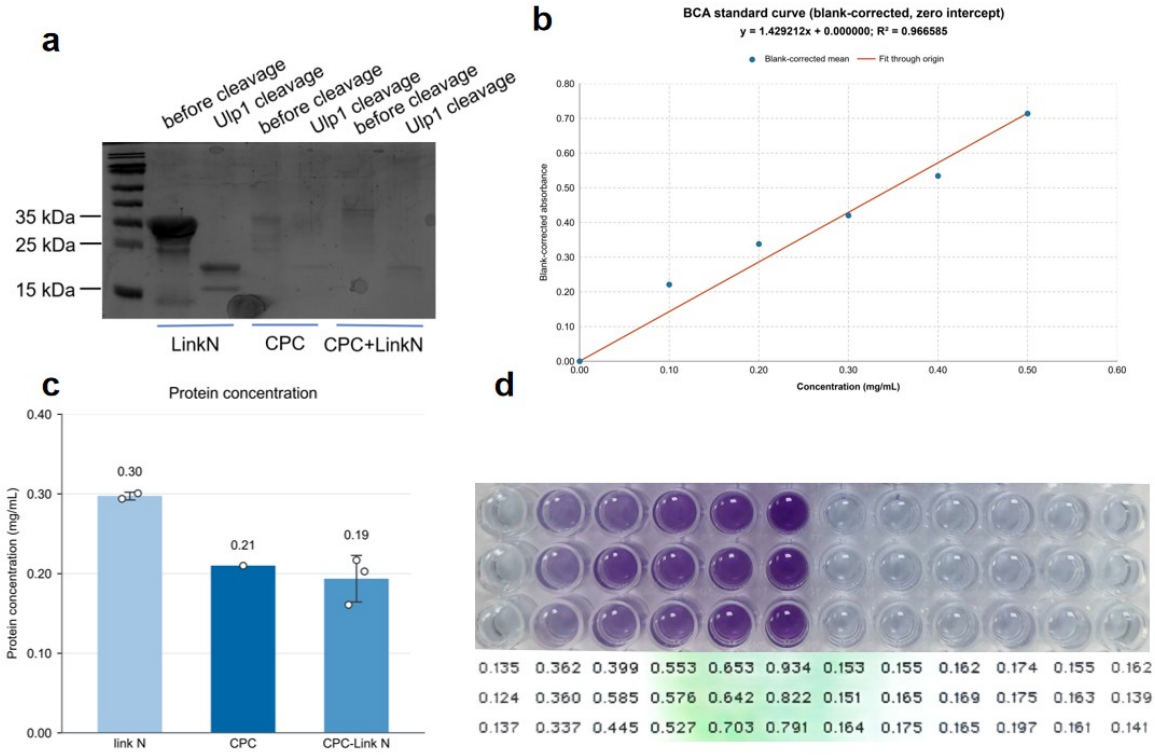


Figure 6. ULP1-mediated SUMO cleavage and BCA quantification. SDS-PAGE shows cleavage-associated band changes but also residual uncleaved material; the accompanying concentration chart reports Link-N 0.30 mg/mL, CPC 0.21 mg/mL and CPC-Link-N 0.19 mg/mL for that batch.

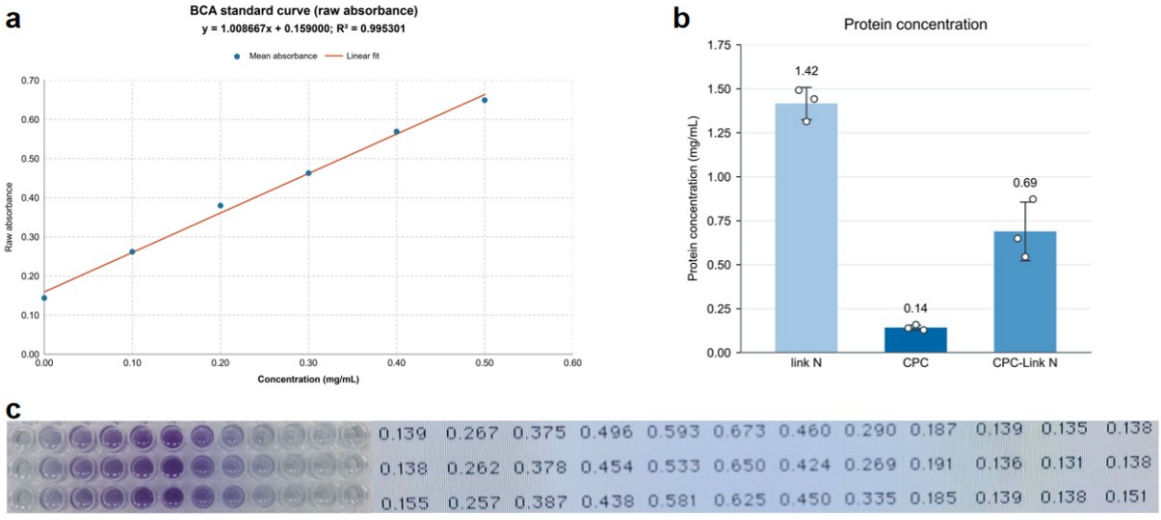


Figure 7. BCA measurement after reverse affinity purification and further ultrafiltration. The summary plot reports Link-N 1.42 mg/mL, CPC 0.14 mg/mL and CPC-Link-N 0.69 mg/mL; compare with the separately dated 15 September record noted above.

8 DogTag-mediated ligation activity check

Date: 11 September 2026

Procedures

1. The DogTag-containing CPC / Link-N constructs were mixed with the GFP-SnoopTag reporter system. SnoopLigase2-containing reactions were prepared alongside corresponding reactions without ligase.
2. SUMO-DogTag was included as a positive-control substrate. The reaction mixtures were analysed by SDS-PAGE to determine whether a higher-molecular-weight product consistent with ligation was formed.

Note. Higher-molecular-weight bands in the approximate 55–70 kDa region were observed in enzyme-containing reactions, whereas corresponding no-ligase controls did not show the same product pattern. This supports the presence of DogTag-mediated ligation activity in the recovered constructs.

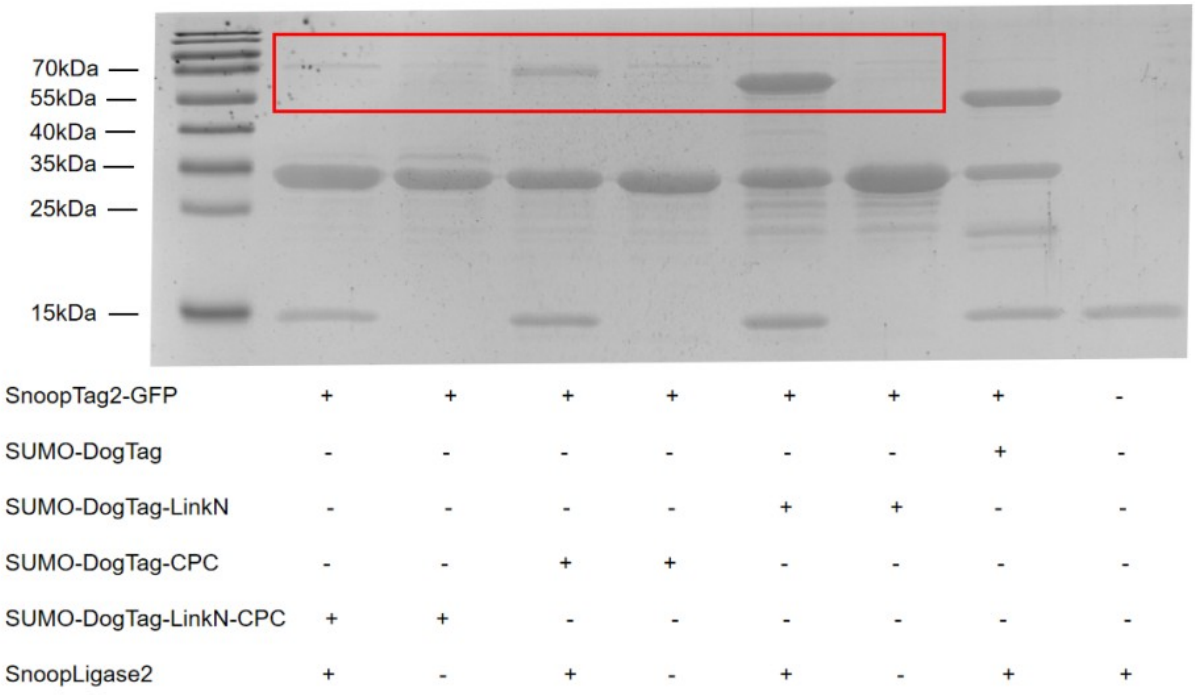


Figure 8. SDS-PAGE assessment of DogTag-mediated ligation to GFP-SnoopTag. Higher-molecular-weight species are marked by the red box; matched reactions without SnoopLigase2 did not show a comparable product pattern.

9 AviTag biotinylation, streptavidin validation and magnetic-bead preparation

Date: 12 and 25 September 2026

Procedures

1. An initial biotinylation and streptavidin-binding attempt was made on 12 September. The recorded ultrafiltration step did not appreciably change the protein concentration, and the subsequent streptavidin SDS-PAGE test was performed without desalting; the result was judged unsatisfactory.
2. The biotinylation workflow was repeated for Link-N, CPC and CPC+Link-N using the BirA method for the AviTag. The processed proteins were desalted to remove free biotin and other small-molecule components before the streptavidin-binding check.
3. For the binding check, the protein samples were incubated with free streptavidin for 30 min and analysed by SDS-PAGE. An unbiotinylated Link-N plus streptavidin control was included for comparison.
4. For the capture assay, streptavidin magnetic beads were washed and incubated overnight at 4 °C with the four recorded sample groups: CK, Link-N, CPC and Link-N+CPC. The bead-bound samples were then used in the CS-A capture workflow.

Note. The unbiotinylated Link-N control retained a visible free-protein band near 15 kDa after incubation with streptavidin, whereas the biotinylated Link-N, CPC and CPC+Link-N samples showed a diminished free-protein signal and higher-molecular-weight species around the 70 kDa region. This supports AviTag biotinylation and streptavidin binding.

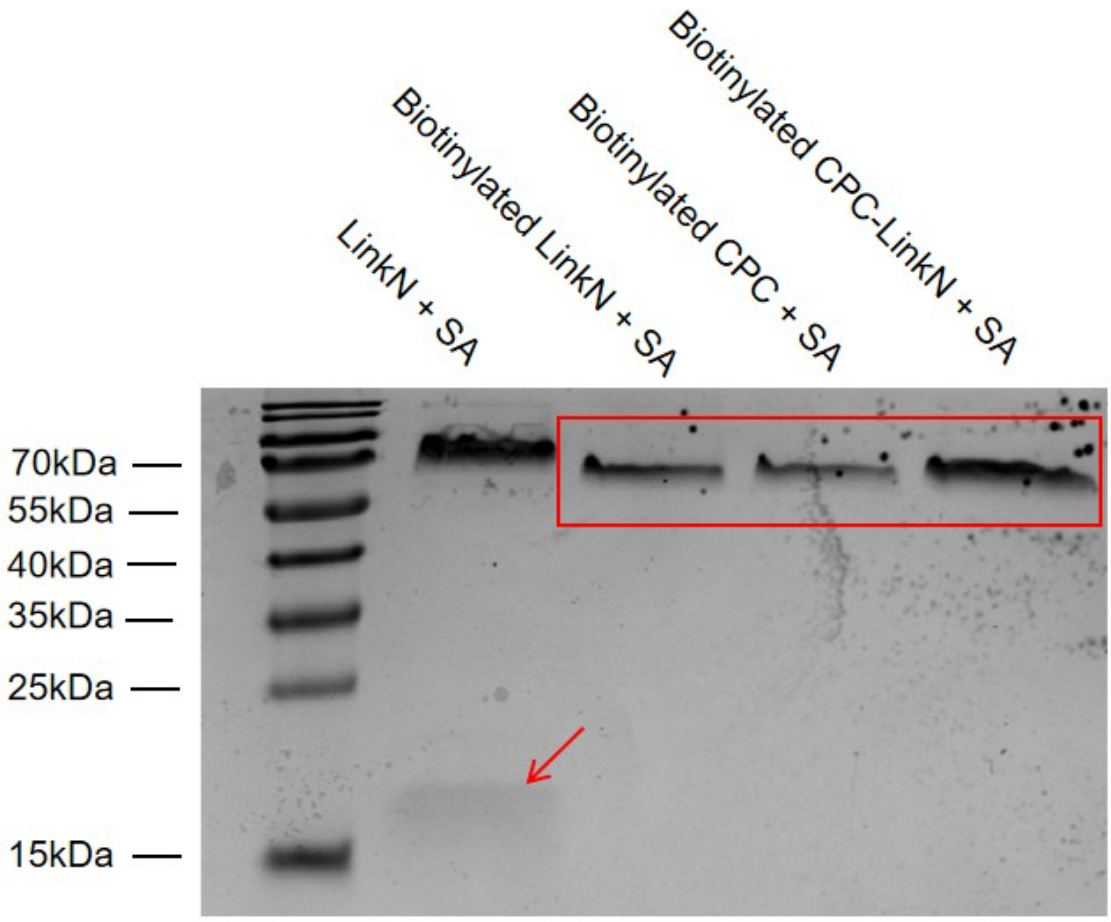


Figure 9. Streptavidin-binding check after AviTag biotinylation. The unbiotinylated Link-N control retains a free-protein band, whereas the biotinylated samples show higher-molecular-weight streptavidin-associated species.

10 DMMB quantification of residual CS-A after magnetic-bead capture

Date: 27–28 September 2026

Procedures

1. After overnight incubation of the bead groups with the CS-A substrate, magnetic separation was used to collect the supernatants for DMMB measurement. The four sample labels were CK (beads without immobilised target protein), L (Link-N), C (CPC) and L+C (Link-N+CPC).
2. The recorded assay response was calculated as OD525 − OD600. A CS-A standard series spanning 0–40 μg/mL was used to fit the standard curve $y = 0.01309006x + 0.04844854$, with $R^2 = 0.8888$. The sample concentrations were calculated from the common standard curve and corrected using the recorded four-fold dilution factor.
3. Three paired readings were reported for each group. The concentration values in the Results summary are reproduced below as recorded; R1–R3 are the three corresponding measurements.

Replicate	CK	L	C	L+C
R1	162.43	162.43	104.37	131.87
R2	116.59	113.54	95.20	95.20
R3	208.27	226.60	168.54	180.76

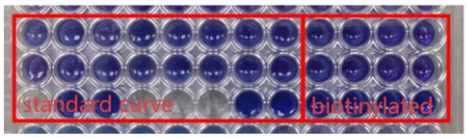
Reported unit: μg/mL after applying the four-fold dilution correction.

Note. Across the three paired readouts, C and L+C were lower than CK, while L did not show a consistent decrease. This direction is consistent with reduced residual CS-A in the supernatant after incubation with the CPC-containing bead groups. It is a preliminary capture-associated result: it does not independently distinguish specific GAG capture from nonspecific adsorption or other losses in the bead workflow.

a

Original concentration (μg/mL)	CK	L	C	L+C
R1	162.43	162.43	104.37	131.87
R2	116.59	113.54	95.20	95.20
R3	208.27	226.60	168.54	180.76

b



c

GAG standard (μg/mL)	R1	R2	R3	Mean ± SD	n
0	0.000	0.020	0.000	0.007 ± 0.012	3
1	0.030	0.020	-0.020	0.010 ± 0.026	3
2.5	0.170	0.090	0.290	0.183 ± 0.101	3
5	0.110	0.110	NA	0.110 ± 0.000	2
7.5	0.060	0.130	NA	0.095 ± 0.049	2
10	0.240	0.110	NA	0.175 ± 0.092	2
20	0.400	0.240	0.560	0.400 ± 0.160	3
40	0.580	0.400	0.620	0.533 ± 0.117	3

d

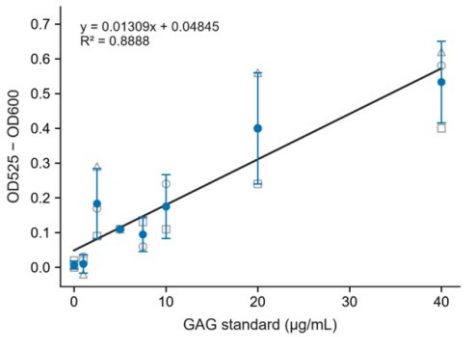


Figure 10. DMMB readout for residual CS-A. Panel A summarises the reported group concentrations; panels C–D show the standard data and fitted calibration curve (y = 0.01309x + 0.04845, R^2 = 0.8888).

Statistical summary reported in the Results document

Comparison	Alternative hypothesis	Mean paired difference (μg/mL)	Raw p	Holm-adjusted p	Decision (α = 0.05)
CPC vs CK	Residual GAG < CK	−39.73	0.0314	0.0314	Reject H0
CPC–Link-N vs CK	Residual GAG < CK	−26.49	0.0053	0.0106	Reject H0

Mean paired difference = treatment − matched CK. Negative values indicate lower residual GAG than CK.
One-sided paired t-tests (n = 3 matched measurements); Holm correction across two prespecified comparisons.
Results are exploratory because all measurements were obtained within one experiment.

Figure 11. Statistical comparison reported in the Results document. The p-values, sidedness label and their consistency with the displayed raw concentrations should be checked before public release.

Validation boundary at the end of this notebook

The work established an expression route for the revised CPC-containing constructs, recovered proteins suitable for downstream assays, detected DogTag-mediated ligation activity, and set up an AviTag–streptavidin magnetic-bead workflow for measuring residual CS-A. The optimised CS-A agarose assay did not yield a clear mobility shift, whereas the bead/DMMB assay showed a preliminary decrease in residual CS-A in the CPC and CPC-Link-N-labelled groups. The available experiments do not directly validate the cartilage-related biological activity of Link-N, and the bead assay does not yet establish a fully background-corrected binding constant or definitive capture efficiency.